AgriSoko Pocket G11 - Project Report

Group G11 | Version 1.0.0 | Repository cxnisme/agrisoko-pocket-g11 | 23 October 2026

Project overview

AgriSoko Pocket G11 is an Expo/React Native Android application paired with a FastAPI member lookup service. It supports task-based member lookup, explainable profile matching, and persistent offline fallback. The directory presents GitHub usernames and assigned task areas.

Team leadership and authorship

This is our group project, led by Christian (cxnisme) as group leader. Christian integrated the mobile app, matching logic, API, and offline behavior with assistance from an AI coding assistant for implementation, focused tests, review, and documentation.

System design

The mobile client uses typed bottom-tab navigation, query validation, request timeouts, source badges, and AsyncStorage caching for the directory, profiles, and successful queries. When the API cannot be reached, bundled task records and local token matching provide offline lookup. The API exposes health, model-info, member-directory, profile, and ask endpoints. Matching uses deterministic TF-IDF/cosine-style term weighting with exact name and alias boosts plus confidence and margin checks.

Recorded automated validation

Area	Recorded check	Result
Backend	python -m pytest -q; rerun 23 Oct	13 passed
Mobile	npm test -- --runInBand; rerun 23 Oct	7 passed
Mobile	npm run typecheck; rerun 23 Oct	Passed
Expo	npx expo install --check	Dependencies up to date
Expo	npx expo-doctor	18/18 checks passed
Android bundle	npx expo export --platform android	Succeeded

Production APK and release

EAS build 51746a29-700a-438c-b3eb-3357ddd7fd9a succeeded using app version 1.0.0 (version code 1), Expo SDK 53.0.0, and source revision 606779e. The remote v1.0.0 tag points to that source revision. GitHub release v1.0.0 is published with build/AgriSokoPocket_G11.apk.

The APK SHA-256 is AD64418C85B1B7AA56DCF3FC5F5DBF9AFA579038E4DCCC64C2343DEFA398F2A8. Archive inspection found 1,065 entries, including AndroidManifest.xml, classes.dex, resources.arsc, and native libraries.

Known limitation and remaining evidence

The APK bundle uses the Android emulator fallback http://10.0.2.2:8000; a hosted API URL was not embedded. Live API use on physical devices requires API deployment, EAS production EXPO_PUBLIC_API_URL configuration, and a replacement APK. No physical-device test, demo recording, external user feedback, member profile sign-off, or matcher evaluation artifact is included.

Submission package

The COMBO_A3_G11 package contains the source, report, contribution and test records, feedback status, AI-use disclosure, and the published APK. The demo recording is not available and is not represented as completed.